

REBECCA M. M. HICKE

PHD CANDIDATE, COMPUTER SCIENCE, CORNELL UNIVERSITY

Telephone: (847) 800-1183

Email: rmh327@cornell.edu

Website: <https://rmatouschekh.github.io>

EDUCATION

Cornell University, Ph.D. Candidate, Computer Science (expected) May 2027
Minor: Information Science Ithaca, NY
Committee: David Mimno (chair), Lillian Lee, Jon Kleinberg, Ross Kristensen-McLachlan
Thesis areas: Computational literary studies, natural language processing (NLP)

Cornell University, M.S., Computer Science October 2024
Minor: Information Science Ithaca, NY

Carleton College, B.A. *cum laude*, Computer Science and English June 2022
Minor: Digital Arts and Humanities Northfield, MN
Advisors: Eric Alexander and George Shuffelton
Received distinction in both majors

RESEARCH & INDUSTRY EXPERIENCE

Graduate Research Assistant, Cornell University August 2022 – present
Advisor: David Mimno Ithaca, NY
Areas: Computational literary studies, NLP

AI Research Scientist, AIIL Group, Microsoft Research November 2025 – January 2026
Advisor: Kiran Tomlinson Remote
Areas: Human-AI interaction, NLP

Research Intern, AIIL Group, Microsoft Research May 2025 – August 2025
Advisor: Kiran Tomlinson Redmond, WA
Areas: Human-AI interaction, NLP

Graduate Research Assistant, Aarhus University May 2024 – August 2024
Advisor: Ross Deans Kristensen-McLachlan Aarhus, Denmark
Areas: Computational literary studies, NLP

PiTech Research Fellow, Cornell University June 2023 – August 2023
Host Organization: Good Call NYC Remote
Areas: Data visualization, data analytics

Research Scientist, Authors AI May 2023 – August 2023
Areas: NLP Remote

Undergraduate Research Assistant, Carleton College June 2020 – May 2022
Advisor: Eric Alexander Northfield, MN
Areas: Data visualization, Human-computer interaction (HCI), digital humanities

Undergraduate Research Assistant, UC Berkeley June 2021 – March 2022
REU – Summer Undergraduate Program in Engineering Research at Berkeley Remote
Advisors: Sarah Chasins and Aditya Parameswaran
Areas: HCI, NLP

TEACHING EXPERIENCE

Instructor of Record, Cornell University June 2026 – August 2026
INFO 1100: Introduction to Programming
Designed course and materials (available on my website)

Research Group Instructor and Mentor, Cornell University Spring 2024 – Spring 2025
Content Analysis and Large Language Model Assisted Narrative Research
Co-designed and directed research course, mentored 8 students

Guest Lecturer

Text Mining History and Literature, Cornell University August 2026
taught by David Mimno
NLP for Cultural Analytics, University of Colorado Boulder September 2025
taught by Maria Antoniak
Natural Language Processing, Carleton College October 2024
taught by Eric Alexander
How LLMs Work: Their Potential and Limitations, Cornell University October 2024
taught by David Mimno

Graduate Teaching Assistant, Cornell University August 2023 – present
INFO 3350: Text Mining History and Literature (1 term)
CS/INFO 4300: Language and Information (1 term)
Received Outstanding TAing Award
CS 2110: Object-Oriented Programming and Data Structures (1 term)
CS/INFO 3300: Data-Driven Web Applications (1 term)

Undergraduate Teaching Assistant, Carleton College March 2020 – June 2022
CS 111: Introduction to Computer Science (1 term)
CS 201: Data Structures (4 terms)
CS 202: Math of Computer Science (2 terms)

PUBLICATIONS

* Represents undergraduate mentees at the time of publication

Peer-Reviewed Conference and Journal Publications

Hicke, Rebecca M. M. and Kiran Tomlinson. **Adopt $\neq$ Adapt: Longitudinal Analyses of LLM Conversations in the Wild**. *Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.

- Hicke, Rebecca M. M., Brian W. Haggard, Mia Ferrante,* Rayhan Khanna,* and David Mimno. **Are You There God? Lightweight Narrative Annotation of Christian Fiction with LMs.** *Conference on Computational Humanities Research (CHR 2025).*
- Hicke, Rebecca M. M., Yuri Bizzoni, Pascale Feldkamp, and Ross Deans Kristensen-McLachlan. **Says Who? Effective Zero-Shot Annotation of Focalization.** *Conference on Computational Humanities Research (CHR 2025).*
- Hicke, Rebecca M. M., Sil Hamilton, and David Mimno. **The Zero Body Problem: Probing LLM Use of Sensory Language.** *Conference on Language Modeling (COLM 2025).*
- Hicke, Rebecca M. M. and David Mimno. **Looking for the Inner Music: Probing LLMs' Understanding of Literary Style.** *Expanding the Toolkit: Large Language Models in Humanities Research in Computational Humanities Research (2025).*
- Hicke, Rebecca M. M. and Ross Deans Kristensen-McLachlan. **SCIENCE IS EXPLORATION: Computational Frontiers for Conceptual Metaphor Theory.** *Conference on Computational Humanities Research (CHR 2024).*
Finalist – Best Short Paper Award
- Kristensen-McLachlan, Ross Deans, Rebecca M. M. Hicke, Márton Kardos, and Mette Thunø. **Context is Key(NMF): Modelling Topical Information Dynamics in Chinese Diaspora Media.** *Conference on Computational Humanities Research (CHR 2024).*
- Hicke, Rebecca M. M. and David Mimno. **T5 meets Tybalt: Author Attribution in Early Modern English Drama using Large Language Models.** *Conference on Computational Humanities Research (CHR 2023).*

Book Chapters

- Hicke, Rebecca M. M. and David Mimno. **'Let every word weigh heavy of her worth': Examining How Women Enact Power in Shakespeare's Comedies through Interactive Speech Pattern Visualizations.** In *Computational Drama Analysis: Reflecting Methods and Interpretations* (2024). De Gruyter.

Workshop Publications

- Hamilton, Sil, Rebecca M. M. Hicke, Mia Ferrante,* David Mimno, and Matthew Wilkens. **Too Long, Didn't Model: Decomposing LLM Long-Context Understanding with Novels.** *Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature (LaTeCH-CLfL 2026).*
- Hamilton, Sil, Rebecca M. M. Hicke, Matthew Wilkens, and David Mimno. **A City of Millions: Mapping Literary Social Networks at Scale.** *International Conference on Natural Language Processing for Digital Humanities (NLP4DH 2025).*
- Hicke, Rebecca M. M. and David Mimno. **[Lions: 1] and [Tigers: 2] and [Bears: 3], Oh My! Literary Coreference Annotation with LLMs.** *Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature (LaTeCH-CLfL 2024).*

Hicke, Rebecca M. M., Maanya Goenka, and Eric Alexander. **Word Clouds in the Wild**. *IEEE Workshop on Visualization for the Digital Humanities (Vis4DH 2022)*.

Under Review

Hicke, Rebecca M. M., Sil Hamilton, David Mimno, and Ross Deans Kristensen-McLachlan. **The Plot Thins: Uniformity and Linearity in Literary Summaries**. *Under review at CHR 2027*.

Bax, Axel, Sanghoon Oh, Rebecca M. M. Hicke, Matthew Wilkens, and David Mimno. **Judging a Cover by its Book**. *Under review at CHR 2027*.

SERVICE

Workshop Organizer: NLP for the Digital Humanities [NLP4DH] (2025)

Session Chair: CHR (2023)

Reviewer: LaTeCH-CLfL (2024, 2025, 2026), DH (2024, 2025, 2026), CHR (2023, 2026), CHR Journal (2025), Open Research Europe (2023), Journal of Computational Literary Studies (2023)

Mentoring and Outreach:

ACSU Research Night (2023, 2024, 2025)

Recruited graduate presenters for outreach getting students interested in research

Student-Applicant Support Program (2022, 2023, 2024, 2025)

Provided feedback on application materials for applicants to Cornell CS PhDs

New Visions in Engineering (2022-2023, Coordinator: 2023-2024)

Mentored high school students interested in futures in engineering + CS research

Bowers Undergraduate Research Experience (BURE) (2023)

Mentored and organized social events for undergraduate research assistants

Program Assistant, Carleton College (June 2022 – August 2022)

Mentored and supervised students in History of Computing Study-Abroad Program

Residential and Program Assistant at UT Austin (2018 – 2020)

Academy for Women – formerly First Bytes (2 sessions)

Academy for All – formerly Code Longhorn (1 session)

Academy for Robotics (2 sessions)

Academy for Cybersecurity (1 session)

AWARDS & FELLOWSHIPS

Future Leaders in Liberal Arts AI Education Travel Grant, <i>Harvey Mudd College</i>	May 2026
Foundational AI PhD Fellowship, <i>Cornell University</i>	January 2026 – May 2026
Hopcroft Fellowship, <i>Cornell University</i>	2022 – 2023
Laurence McKinley Gould Prize in Natural Science, <i>Carleton College</i>	May 2022
David John Field Prize, <i>Carleton College</i>	May 2022

The David Pollatsek '96 Prize in Computer Science, <i>Carleton College</i>	May 2022
<i>Distinction</i> – Computer Science Senior Thesis, <i>Carleton College</i>	May 2022
<i>Distinction</i> – English Senior Thesis, <i>Carleton College</i>	April 2022
Thorsen Scholarship, <i>Carleton College</i>	2020 – 2022
Larson International Fellowship, <i>Carleton College</i>	2021

CERTIFICATIONS & SUPPLEMENTAL TRAINING

(Re)Imagining Liberal Arts & STEM Education in the Age of GenAI, Harvey Mudd College <i>Conference on adapting education in response to the proliferation of AI</i>	May 2026
Early Career Graduate Teaching Cohort, Cornell University <i>Workshop series on developing teaching skills</i>	Spring 2026
Essentials of Teaching Institute, Cornell University <i>Course on developing teaching skills</i>	Fall 2025

PRESENTATIONS

Invited Talks

“Says Who? Effective Zero-Shot Annotation of Focalization,” <i>Quid deinde? New Ideas and Approaches to AI & DH, University of Montreal</i>	April 2026
“How to Read Books with AI: AI and Narrative Analysis,” <i>University of Richmond</i>	March 2026

Conference Talks

“Says Who? Effective Zero-Shot Annotation of Focalization,” <i>CHR 2025</i>	December 2025
“Are You There God? Lightweight Narrative Annotation of Christian Fiction with LMs,” <i>CHR 2025</i>	December 2025
“SCIENCE IS EXPLORATION: Computational Frontiers for Conceptual Metaphor Theory,” <i>CHR 2024</i>	December 2024
“T5 meets Tybalt: Author Attribution in Early Modern English Drama Using Large Language Models,” <i>CHR 2023</i>	December 2023

Workshop Talks

“Word Clouds in the Wild,” Vis4DH @ IEEE Vis	October 2022
“Examining How Women Enact Power in Shakespeare’s Comedies through Interactive Speech Pattern Visualizations,” <i>Workshop on Computational Drama Analysis</i>	September 2022

Poster Presentations

“The Zero Body Problem: Probing LLM Use of Sensory Language,” <i>COLM 2025</i>	October 2025
--	--------------

“[Lions: 1] and [Tigers: 2] and [Bears: 3], Oh My! Literary Coreference with LLMs,” <i>LaTeCH-CLfL 2024</i>	March 2024
“Large Language Models and NER: Better Results with Less Work,” <i>DH 2023</i>	July 2023
“T5 meets Tybalt: Author Attribution in Early Modern English Drama Using Large Language Models,” <i>TADA 2023</i>	November 2023